



Decision Analysis



Introduction to Decision Analysis

-
- Decision-making often involves uncertainty.
 - Previous chapters focused on decisions with known consequences.
 - Decision analysis provides a framework for rational decision-making under uncertainty.

Examples of Uncertain Decision Environments:

1. Introducing a new product (customer reaction etc.)
2. Financial investments (market sectors, economy, interest rates).
3. Government contract bidding (costs, competitors' bids).
4. Agricultural decisions (weather, prices, costs).
5. Oil drilling decisions (likelihood of oil, drilling depth).

Purpose of Decision Analysis

Objective

- Address decision-making in environments under uncertainty.

Tools

- Mathematical models, experimentation, and decision trees.

Outcome

- Rational and informed decisions.

Section 16.1: A Prototype Example

Example

The Goferbroke Company Problem

- **Scenario:**

- The company owns land that may contain oil.
- A consulting geologist estimates a **1 in 4 chance** of finding oil.
- Another oil company offers to buy the land for **\$90,000**.
- Goferbroke considers drilling for oil at a cost of **\$100,000**.

Potential Outcomes:

- **If oil is found:** Revenue = 800,000; *Profit* = 700,000.
- **If land is dry:** Loss = \$100,000 (drilling cost).

Example (Cont...)

Decision Alternatives:

1. Drill for oil:

1. Potential profit: \$700,000 (if oil is found).
2. Potential loss: \$100,000 (if land is dry).

2. Sell the land:

1. Guaranteed profit: \$90

■ **TABLE 16.1** Prospective profits for the Goferbroke Company

Alternative	Status of Land	Payoff	
		Oil	Dry
Drill for oil		\$700,000	-\$100,000
Sell the land		\$ 90,000	\$ 90,000
Chance of status		1 in 4	3 in 4

Some key Questions

- Should Goferbroke **drill for oil** or **sell the land**?
- Is it worth conducting a **seismic survey** to reduce uncertainty?
- How does the company's **financial situation** affect the decision?

Next steps

- **Section 16.2:** Decision making without experimentation.
- **Section 16.3:** Decision making with experimentation.
- **Section 16.4:** Decision trees for analyzing sequential decisions.

16. 2. Decision Making Without Experimentation.

Introduction to Decision Making without Experimentation

- Decision making under uncertainty where outcomes are not tested beforehand.
- Focuses on choosing alternatives based on available data (no experimentation).
- Key question: How to make rational decisions when consequences are uncertain?

General Framework for Decision Making

Key Components:

1. Decision Alternatives:

- Set of feasible actions the decision maker can choose.

2. States of Nature:

- Possible situations that affect the outcome (e.g., oil or dry land).

3. Payoffs:

- Quantitative measure of the outcome's value (e.g., profit or loss).

Decision Analysis Framework

Steps:

1. Choose a decision alternative.
2. Nature selects a state of nature.
3. Payoff is determined based on the combination of decision and state of nature.
4. Use payoff table to find the optimal decision based on a chosen criterion.

Analogy to Game Theory:

- Decision maker vs. nature (**nature is a passive player** and the decision maker is player 1).

Role of Prior Probabilities

- **Definition:** Probabilities assigned to states of nature based on available information.
- Decision maker uses prior probabilities to estimate likelihood of states of nature.
- Often subjective, based on experience or intuition.
- **Example (Goferbroke):**
 - Probability of oil: 0.25.
 - Probability of dry land: 0.75.

■ **TABLE 16.2** Payoff table for the decision analysis formulation of the first Goferbroke Co. problem

Alternative	State of Nature	
	Oil	Dry
1. Drill for oil	700	-100
2. Sell the land	90	90
Prior probability	0.25	0.75

Decision Criteria

Three Criteria for Decision Making:

1. Maximin Payoff Criterion:

- i. Focuses on the worst-case scenario.
- ii. Choose the alternative with the best worst-case payoff.

2. Maximum Likelihood Criterion:

- i. Focuses on the most likely state of nature.
- ii. Choose the alternative with the best payoff for the most likely state.

3. Bayes' Decision Rule:

- i. Uses expected payoffs based on prior probabilities.
- ii. Choose the alternative with the highest expected payoff.

1. Maximum Payoff Criterion

Maximin Payoff Criterion

Steps:

1. Identify the minimum payoff for each alternative.
2. Choose the alternative with the maximum of these minimum payoffs.

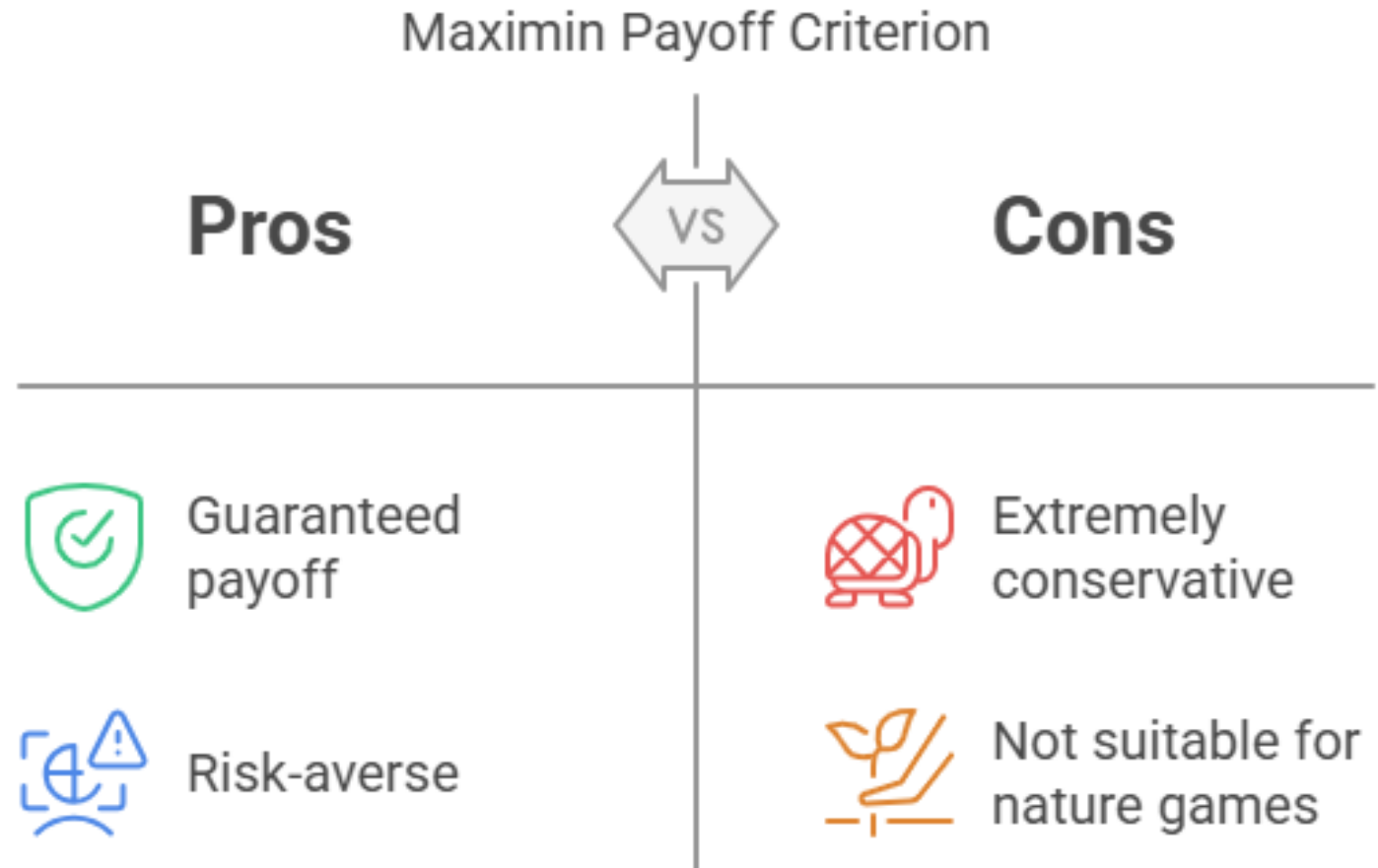
■ **TABLE 16.3** Application of the maximin payoff criterion to the first Goferbroke Co. problem

Alternative	State of Nature		Minimum
	Oil	Dry	
1. Drill for oil	700	−100	−100
2. Sell the land	90	90	90 ← Maximin value
Prior probability	0.25	0.75	

Maximin Payoff Criterion

- **Drill for oil:** Minimum payoff = -\$100,000.
 - **Sell the land:** Minimum payoff = \$90,000.
 - **Decision:** Sell the land (maximin value = \$90,000) – A pessimistic approach
-
- This rationale for this decision criterion is that it provides the best guaranteed of the payoff, regardless of what the true state of nature turns out to be.
 - However, this criterion is not often used in the games against nature as it is extremely conservative criterion as nature is not a conscious opponent

Maximin Payoff Criterion



2. Maximum Likelihood Criterion

Maximum Likelihood Criterion

Steps:

1. Identify the most likely state of nature.
2. Choose the alternative with the best payoff for that state.

Application to Goferbroke:

- Most likely state: Dry land (probability = 0.75)
- **Drill for oil:** Payoff = -\$100,000.
- **Sell the land:** Payoff = \$90,000.
- **Decision:** Sell the land.



Choose the option with the best payoff.

Decision – Sell the land

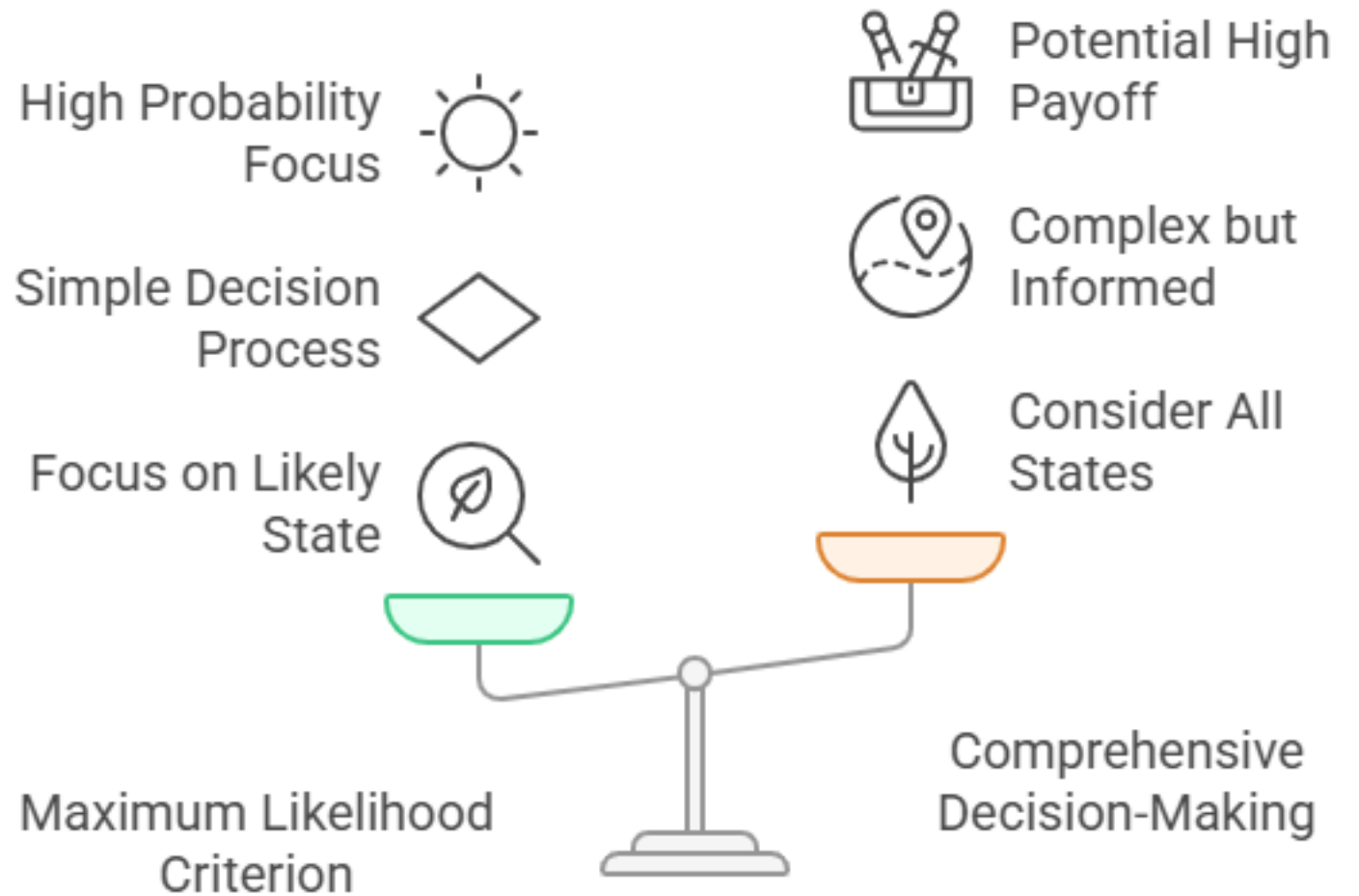
■ **TABLE 16.4** Application of the maximum likelihood criterion to the first Goferbroke Co. problem

Alternative	State of Nature		
	Oil	Dry	
1. Drill for oil	700	-100	-100
2. Sell the land	90	90	90 ← Maximum in this column
Prior probability	0.25	0.75	
		↑ Maximum	



Choose the option with the best payoff.

Drawback of Maximum Likelihood Criterion



Balancing Decision-Making Approaches

3. Bayes' Decision Rule

Bayes' Decision Rule (Choose the alternative with the maximum expected payoff)

1. Calculate the expected payoff for each alternative.
2. Choose the alternative with the highest expected payoff.

Application to Goferbroke

- **Drill for oil:**
 - Expected payoff = (Probability of oil × Payoff if oil) + (Probability of dry × Payoff if dry).
 - Expected payoff = $(0.25 \times 700) + (0.75 \times (-100)) = \$100.$
- **Sell the land:**
 - Expected payoff = (Probability of oil × Payoff if oil) + (Probability of dry × Payoff if dry).
 - Expected payoff = $(0.25 \times 90) + (0.75 \times 90) = \$90.$
- **Decision:** Since $100 > 90$, drill for oil (higher expected payoff).

Using Bayes' Rule

Using Bayes' decision rule

Pros



Cons



Incorporates all information



Considers payoffs



Reasonable probability estimates



Improves with experimentation



Better decisions



Subjective probability estimates



Uncertainty in predictions

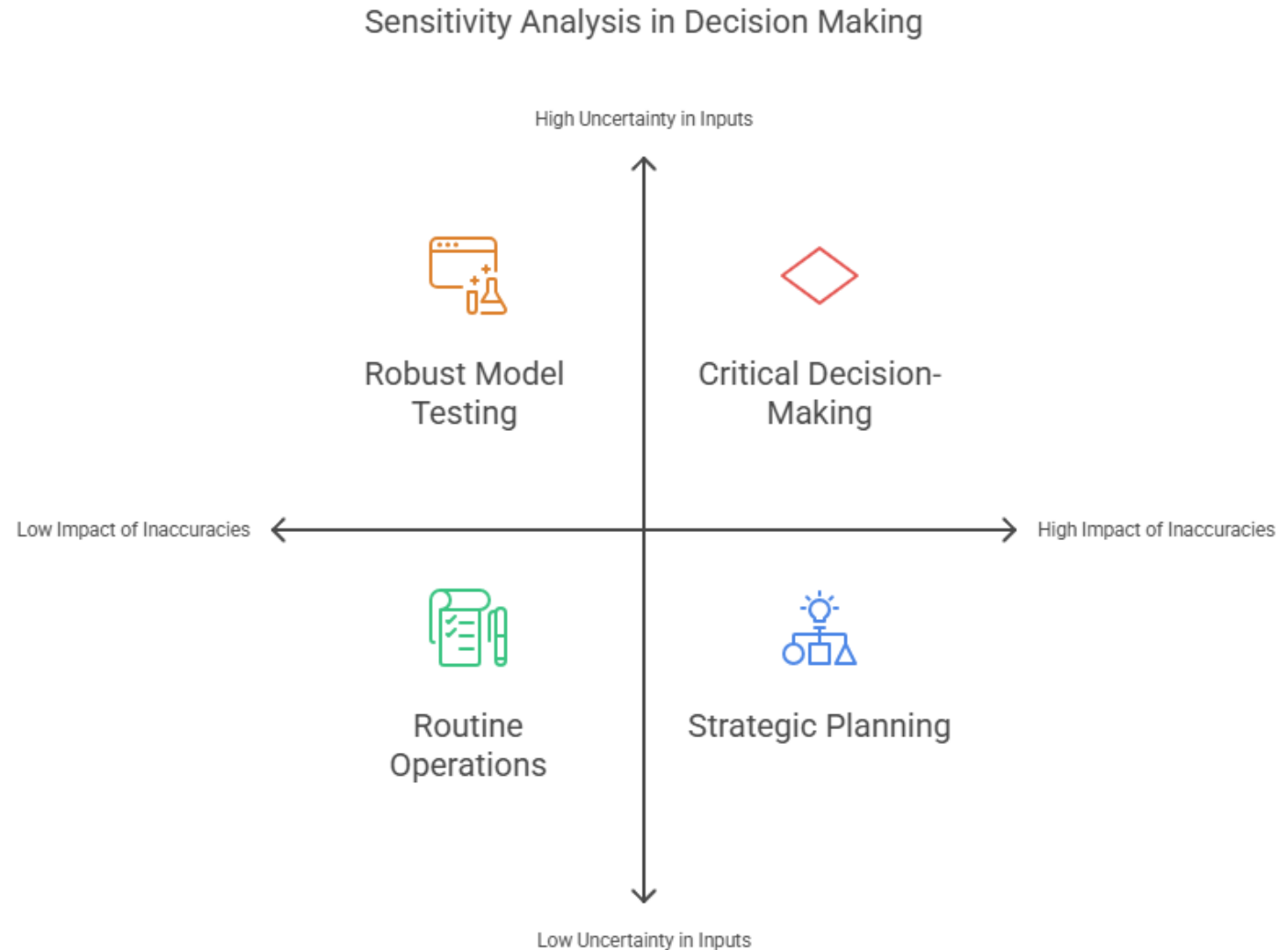


Requires sensitivity analysis

Sensitivity Analysis

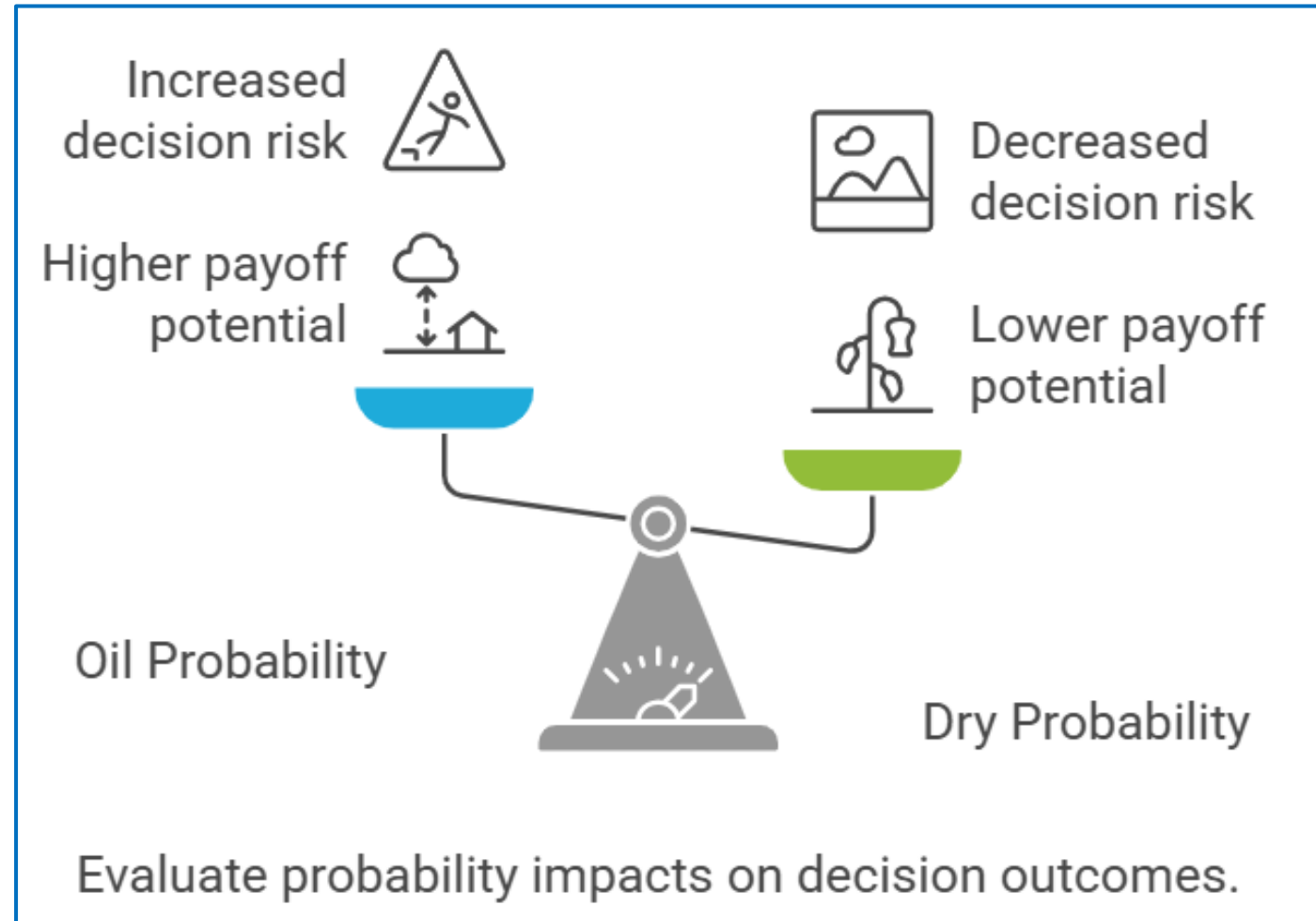
What is Sensitivity Analysis?

- Sensitivity analysis studies the impact of inaccurate numbers in a model.
- Commonly used in operations research to test **model robustness**.
- In decision analysis: Focuses on uncertain inputs like prior probabilities or payoffs.
- **Goal:** Determine how decisions change if estimates are off.



Focus on Prior Probabilities

- In the Goferbroke Co. example, the payoff table (Table 16.2) is the model.
- Most questionable numbers: Prior probabilities (0.25 for oil, 0.75 for dry).
- Why? Probabilities are subjective and hard to predict precisely.
- Sensitivity analysis will test how decisions vary with changes in these probabilities.



Setting up the Analysis

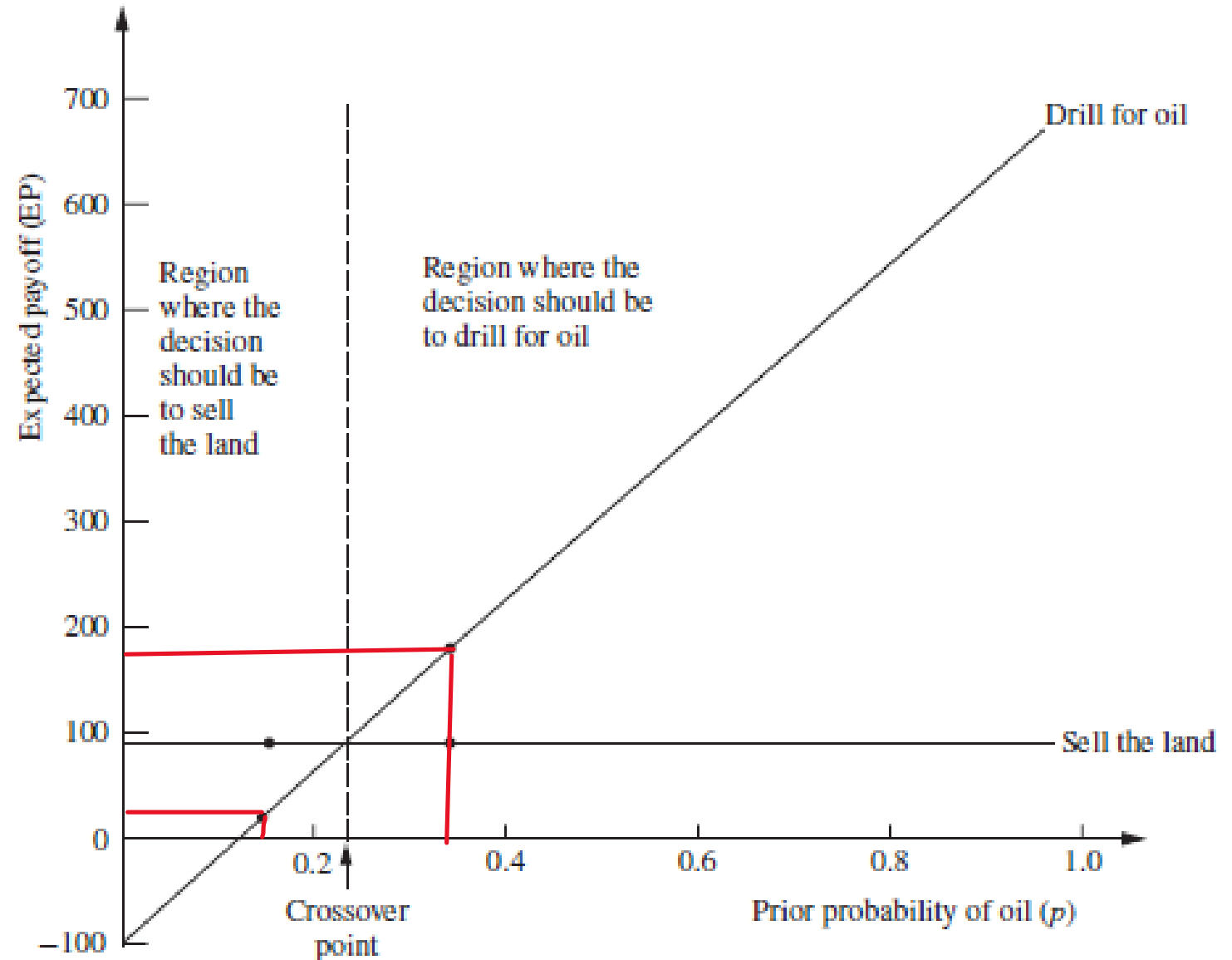
- Sum of prior probabilities = 1 (e.g., p for oil + $(1 - p)$ for dry = 1).
- Goferbroke's estimate: True probability of oil (p) is between 0.15 and 0.35.
 - Dry probability: 0.85 to 0.65.
- Define:
 - p = prior probability of oil.
 - $E[\text{Payoff (drill)}] = 700p - 100(1 - p) = 800p - 100$.
 - $E[\text{Payoff (sell)}] = 90$ (constant, regardless of p).

Expected Payoff Equations

- Drilling: $E[\text{Payoff (drill)}] = 800p - 100$
 - Linear function, increases with p .
- Selling: $E[\text{Payoff (sell)}] = 90$
 - Constant, unaffected by p .
- Sensitivity analysis tests how these payoffs compare as p varies.

Graphical Representation

FIGURE 16.1
Graphical display of how the expected payoff for each decision alternative changes when the prior probability of oil changes for the first Goferbroke Co. problem.



Graphical Representation

- Figure 16.1 plots $E[\text{Payoff (drill)}]$ and $E[\text{Payoff (sell)}]$ vs. p .
 - Drilling: Slanting line ($800p - 100$).
 - Selling: Flat line (90).
- Key points:
 - $p = 0.15$: Drill = 20, Sell = 90 → Sell wins.
 - $p = 0.35$: Drill = 180, Sell = 90 → Drill wins.
- Decision is sensitive to p 's value.

Finding the Crossover Point

- Crossover point: Where $E[\text{Payoff (drill)}] = E[\text{Payoff (sell)}]$.
- Solve: $800p - 100 = 90$
 - $800p = 190$
 - $p = 190 / 800 = 0.2375$.
- Decision rule:
 - If $p < 0.2375 \rightarrow$ Sell the land.
 - If $p > 0.2375 \rightarrow$ Drill for oil.

Interpreting the Results

- At $p = 0.25$ (original estimate), Drill (100) > Sell (90) → Drill is optimal.
- But 0.25 is close to 0.2375 → Decision is borderline.
- Range tested (0.15 to 0.35) shows wide swings:
 - $p = 0.15$: Sell (90) beats Drill (20).
 - $p = 0.35$: Drill (180) beats Sell (90).

Implications of Sensitivity Analysis

- Reveals decision's dependence on precise probability estimates.
- Goferbroke Co. needs a more accurate p (current 0.25 is uncertain).
- Suggests further action (e.g., experimentation like a seismic survey).
- Sensitivity analysis ensures robust decision-making under uncertainty.

How should Goferbroke Co. improve decision-making under uncertainty?

Improve
Probability
Estimates

Enhances accuracy and
reliability of decisions

Conduct
Experimentation

Provides empirical data
to refine estimates

Ensure Robust
Decisions

Strengthens decision-
making framework
under uncertainty

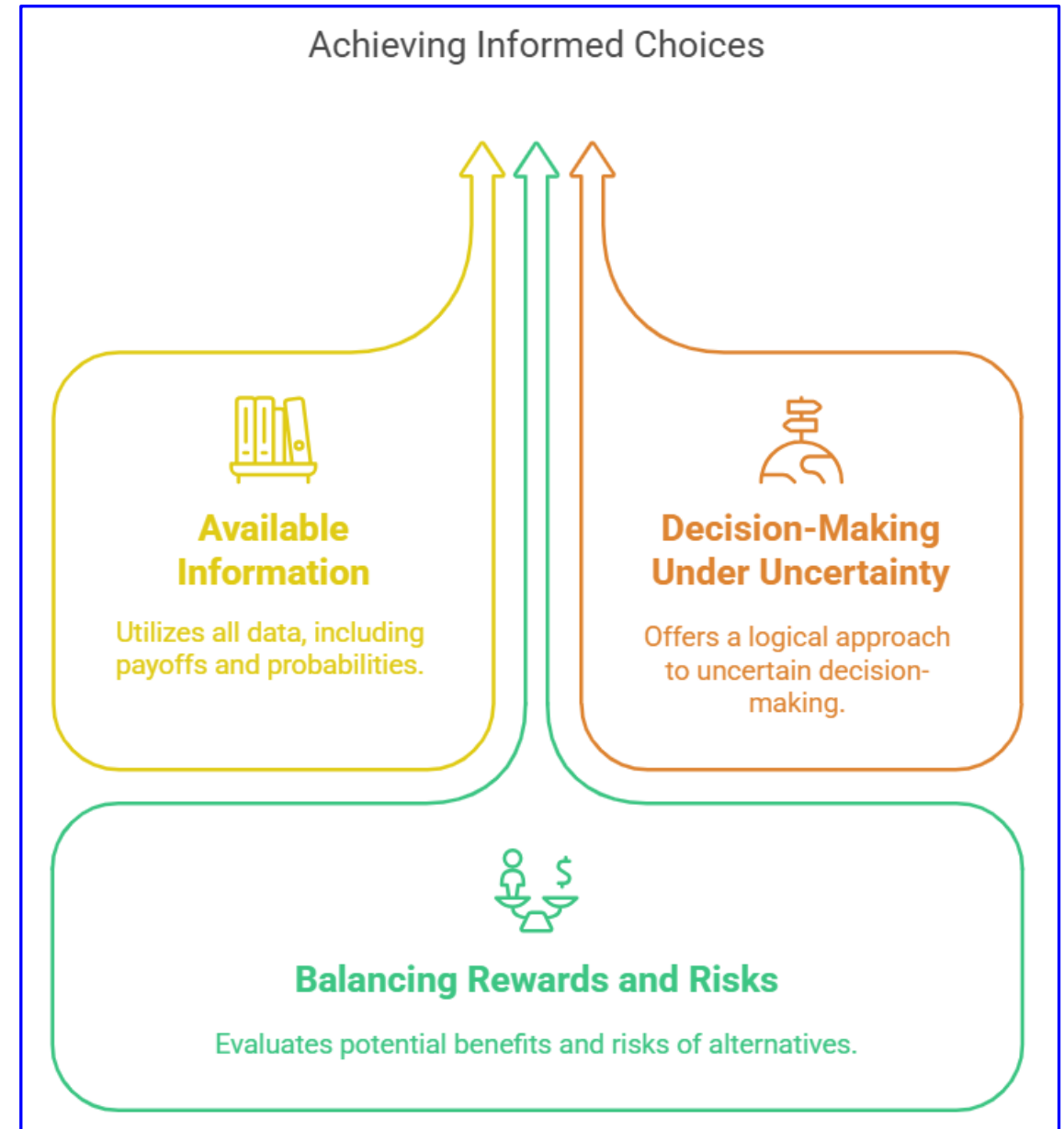


Key Takeaway Points

- Sensitivity analysis tests Bayes' Decision Rule against uncertain probabilities.
- For Goferbroke Co., decision shifts at $p = 0.2375$, highlighting sensitivity.
- Key takeaway: Refine estimates when decisions are close to the crossover.

Why Use Bayes' Decision Rule?

- It incorporates **all available information**, including payoffs and probabilities.
- It provides a **rational basis** for decision-making under uncertainty.
- It balances the potential rewards and risks of each decision alternative.



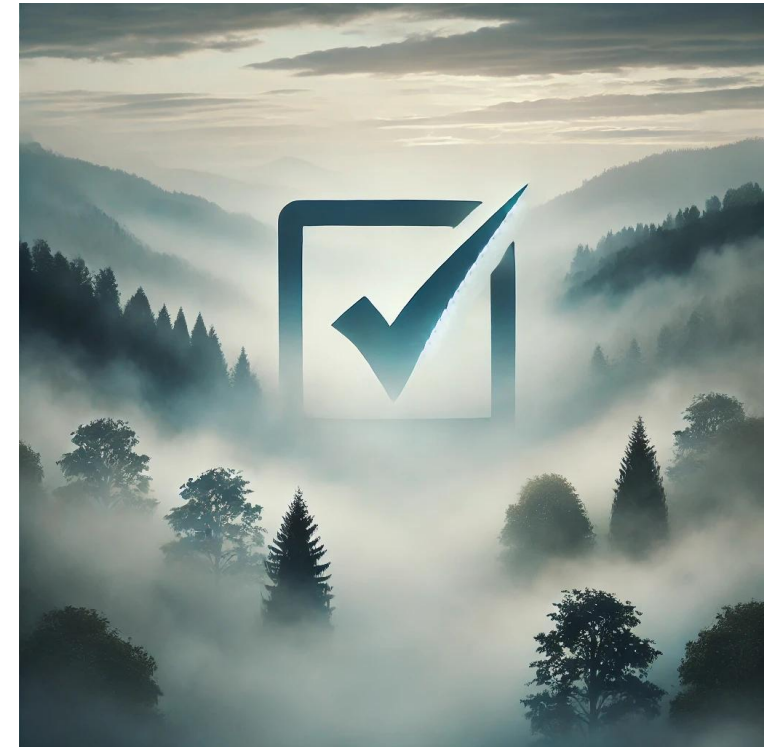
16.3 Decision Making with Experimentation

Improving Decision Analysis through Additional Information

Introduction to Decision Making with Experimentation

- Why Experimentation?

- Decision under uncertainty are risky
- Experimentation helps reduce uncertainty about outcomes.
- It involves gathering additional information (e.g., market research, seismic surveys).
- The goal is to refine **prior probabilities** into **posterior probabilities**.



Understanding Prior and Posterior Probabilities

- Drill: \$ 700K if oil, - \$100K if dry
- Sell: \$90 either way
- **What's a Prior?** Our starting guess about chances.
 - Uncertainty: Oil 25%, Dry 75% (1 in 4 chance of oil).
- **Why Refine?** Testing gives us clues to update our guess.
 - Seismic survey: Favorable (FSS) or Unfavorable (USS).
- **How?** Use test results to get posterior (new) probabilities.

Table 16.1

- Should the company spend \$30K for testing/ experimentation

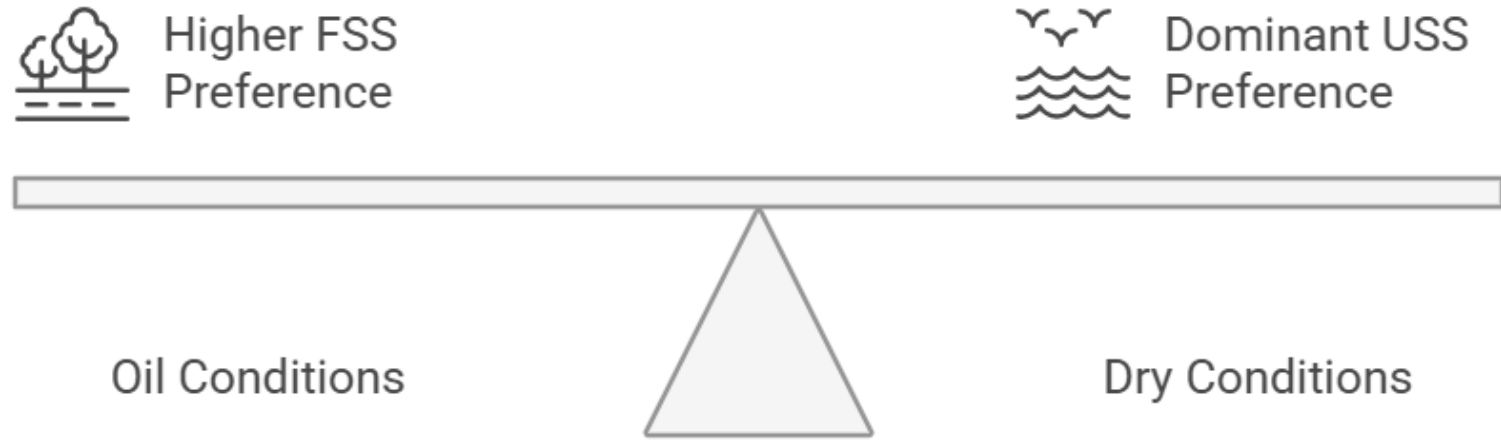
Alternatives	Oil	Dry
Drill	700	-100
Sell	90	90
Chance	0.25	0.75

Changing the Belief

- Through Seismic Survey the company wants to know the oil signs:
- Two possible outcomes
 - FSS (Favorable): Oil seems likely
 - USS (Unfavorable): Oil seems unlikely
- Cost \$30k. Its not perfect but helpful

Survey Reliability

- How good is the survey?
- If Oil: 60% FSS, 40% USS.
- If Dry: 20% FSS, 80% USS.



Strategy Preference by Environmental Conditions

From Prior to Posterior – Refining our Guess

- **Prior:** Starting guess (Oil 0.25, Dry 0.75).
- **Posterior:** New guess after test.
- **Tool:** Bayes' theorem updates probabilities.
 - Start $\rightarrow$ Oil (0.25) $\rightarrow$ FSS (0.6), USS (0.4).
 - Start $\rightarrow$ Dry (0.75) $\rightarrow$ FSS (0.2), USS (0.8).

Bayes' Theorem - The Formula

General Formula:

$$P(State_i | Finding_j) = \frac{P(Finding_j | State_i) \times P(State_i)}{\sum_{k=1}^n P(Finding_j | State_k) \times P(State_k)}$$

In Words:

Posterior = (Likelihood × Prior) ÷ (Sum of all Likelihood × Prior)

“What is the probability that State i is true after we observe Finding j ?”

A bit more explanation

- **1. Prior Probability**

- **$P(\text{State}_i)$** : This means “*How likely was State i before we saw any evidence?*”

- **2. Likelihood**

- **$P(\text{Finding}_j \mid \text{State}_i)$** : This means “*If State i is true, how likely is it that we observe Finding j ?*”

Calculating Posterior Probabilities - USS Case

- For Unfavorable Seismic Soundings (USS):

- $$P(Oil | USS) = \frac{0.4 \times 0.25}{0.4(0.25) + 0.8(0.75)}$$
- $$= \frac{0.1}{0.1 + 0.6} = \frac{0.1}{0.7} = \frac{1}{7} \approx 0.143$$
- $$P(Dry | USS) = 1 - \frac{1}{7} = \frac{6}{7} \approx 0.857$$

Explanation of Posterior Probabilities - USS Case

- **Before the seismic survey:**
 - $P(\text{Oil}) = 0.25$ (prior probability)
 - $P(\text{Dry}) = 0.75$ (prior probability)
- **We conducted the survey and got:**
- **USS** = Unfavorable Seismic Soundings
- **The Question:** *"Given that our survey came back unfavorable, what is the updated probability that the land actually contains oil?"*
- This updated probability is called a **POSTERIOR PROBABILITY**

Calculating Posterior Probabilities - FSS Case

- For Favorable Seismic Soundings (FSS):

- $P(Oil | FSS) = \frac{0.6 \times 0.25}{0.6(0.25) + 0.2(0.75)}$

- $= \frac{0.15}{0.15 + 0.15} = \frac{0.15}{0.3} = \frac{1}{2} = 0.5$

- $P(Dry | FSS) = 1 - \frac{1}{2} = \frac{1}{2} = 0.5$

Expected Payoff with Experimentation

- After obtaining posterior probabilities, calculate the expected payoff for each decision alternative.
- Example: Expected payoff if the survey result is **USS**:

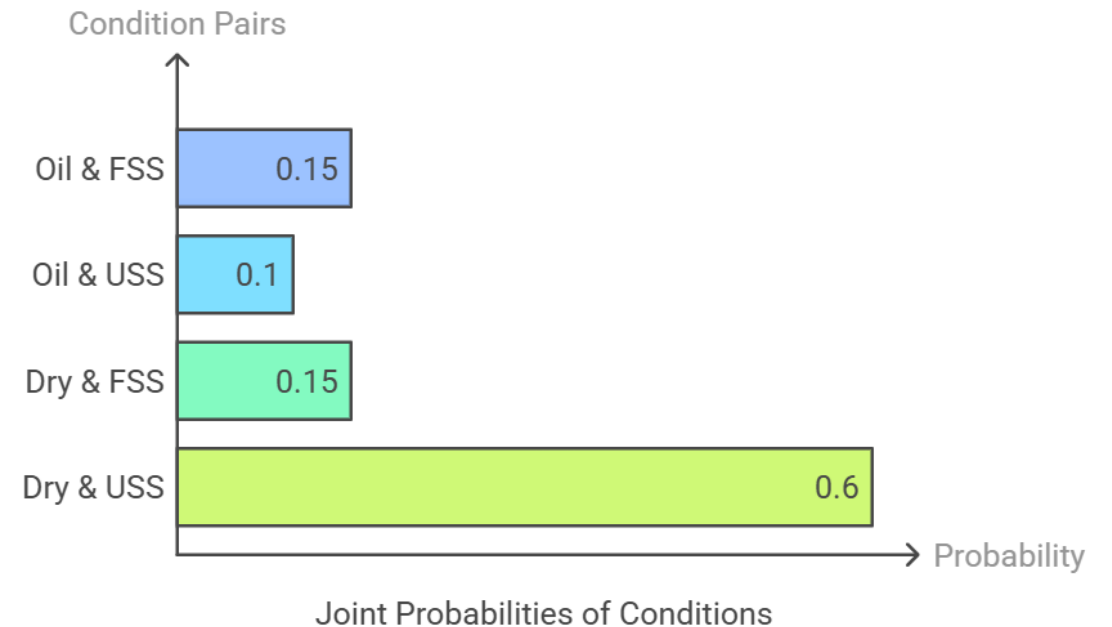
$$E[\text{Payoff (drill)}|\text{USS}] = \frac{1}{7}(700) + \frac{6}{7}(-100) - 30 = -15.7$$

- Expected payoff if the survey result is **FSS**:

$$E[\text{Payoff (drill)}|\text{FSS}] = \frac{1}{2}(700) + \frac{1}{2}(-100) - 30 = 270$$

Joint Probabilities

- Joint Probabilities (initial probability and seismic survey reliability)
- Oil & FSS: $0.25 \times 0.6 = 0.15$.
- Oil & USS: $0.25 \times 0.4 = 0.1$.
- Dry & FSS: $0.75 \times 0.2 = 0.15$.
- Dry & USS: $0.75 \times 0.8 = 0.6$.



Survey Outcomes Odds

- FSS: $0.15 \text{ (Oil)} + 0.15 \text{ (Dry)} = 0.3.$
- USS: $0.1 \text{ (Oil)} + 0.6 \text{ (Dry)} = 0.7.$
- Total: $0.3 + 0.7 = 1.$

Posterior Probabilities

- New Probabilities
 - **FSS**: Oil = $0.15 / 0.3 = 1/2$, Dry = $0.15 / 0.3 = 1/2$
 - 0.5
 - **USS**: Oil = $0.1 / 0.7 = 1/7$, Dry = $0.6 / 0.7 = 6/7$.

Decision Time

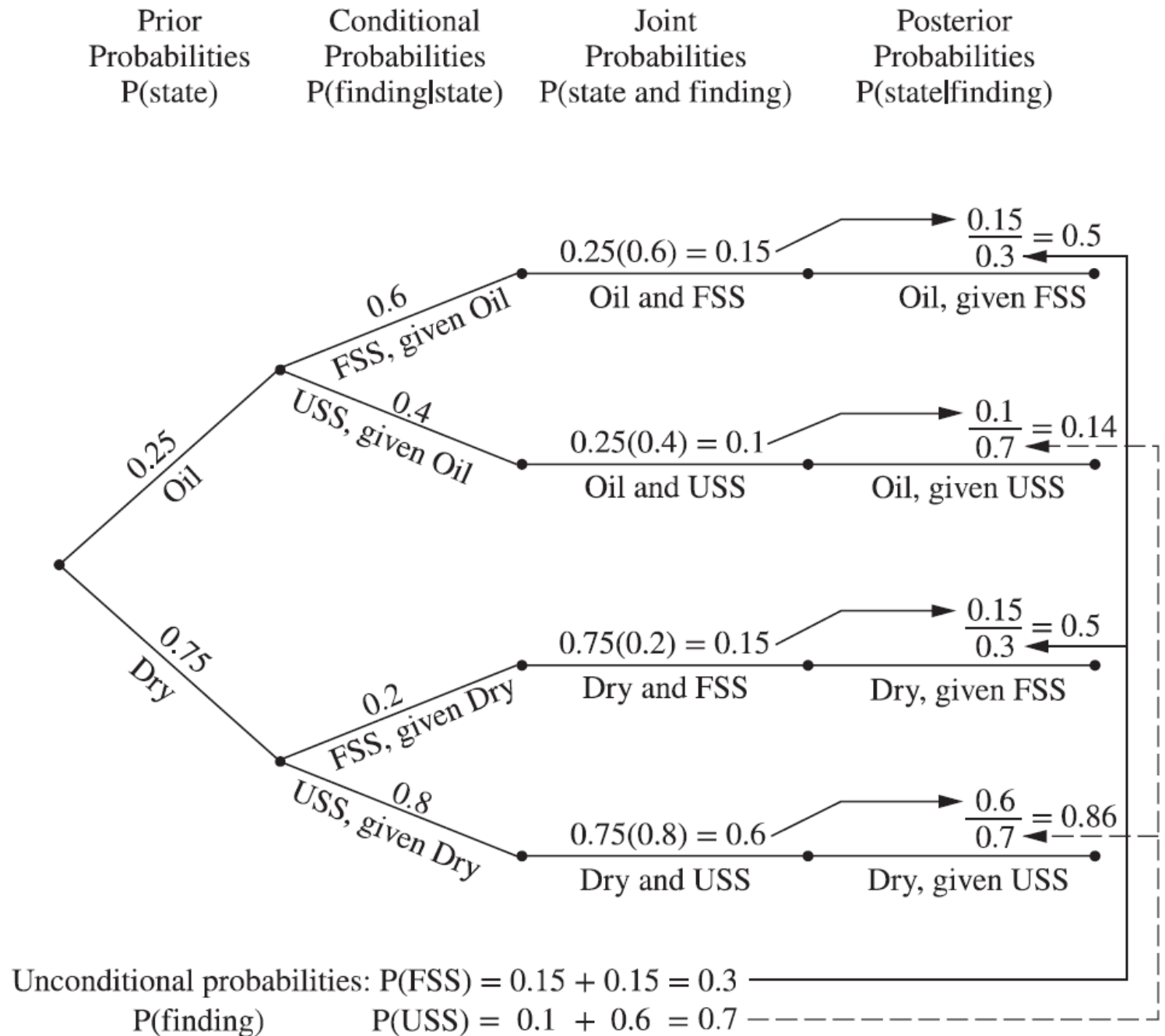
- Using the new Odds
 - **FSS**: Drill ($1/2 \times 700 + 1/2 \times -100 - 30 = 270$) vs. Sell 60 ($60=90-30$)
 - Pick Drill: \$270k.
 - **USS**: Drill ($1/7 \times 700 + 6/7 \times -100 - 30 = -15.7$) vs. Sell (60).
 - Pick Sell: \$60k.

Finding	Best Choice	Payoff
USS	Sell	60
FSS	Drill	270

Probability Tree Diagram

- A probability tree diagram helps visualize the calculation of posterior probabilities.
- **Prior probabilities** are combined with **conditional probabilities** to calculate **joint probabilities**.
- Example: Probability tree for Goferbroke Co. showing prior, conditional, joint, and posterior probabilities.

Probability tree diagram show all probabilities leading to the calculation of each posterior probability of the state of nature given the finding of the seismic survey



Expected Value of Perfect Information (EVPI)

- **Perfect Information** = Knowing with certainty which state will occur
- **Step 1:** Calculate expected payoff WITH perfect information

State	Best Action	Payoff	Prior Prob
Oil	Drill	\$700	0.25
Dry	Sell	\$90	0.75

→ Expected payoff with perfect information = $0.25(700) + 0.75(90)$
= \$242.5

Calculating EVPI

Formula:

EVPI = Expected payoff with perfect information - Expected payoff without experimentation

From before:

- Expected payoff without experimentation = \$100 (drill)
- Expected payoff with perfect information = \$242.5

$$\text{EVPI} = 242.5 - 100 = \$142.5$$

Interpretation: \$142,500 is the maximum we should pay for ANY information

Since survey costs **only \$30,000 (< \$142,500)**, it **MAY be worthwhile**

Expected Value of Experimentation (EVE)

Calculate expected payoff WITH experimentation (excluding cost)

We already have:

- $P(\text{USS}) = 0.7$, Expected payoff if USS = \$90
- $P(\text{FSS}) = 0.3$, Expected payoff if FSS = \$300

Expected payoff with experimentation = $0.7(90) + 0.3(300) = 63 + 90 =$
\$153

Calculating EVE

Formula:

EVE = Expected payoff with experimentation - Expected payoff without experimentation

$$\text{EVE} = 153 - 100 = \$53$$

Interpretation: \$53,000 is the actual expected value of conducting the seismic survey

Since survey costs \$30,000, and $53,000 > 30,000$, the survey is worthwhile

Summary Table - Value of Information

Measure	Value	Interpretation
Expected payoff without experimentation	\$100	Baseline
Expected payoff with experimentation	\$153	After updating probabilities
EVE (Expected Value of Experimentation)	\$53	Net gain from survey (before cost)
Cost of survey	\$30	Actual expense
Net gain after cost	\$23	Survey is profitable
EVPI (Expected value of perfect information)	\$142.5	Upper bound - theoretical maximum

Value of Experimentation

- **Expected Value of Perfect Information (EVPI):**
 - The maximum amount one should be willing to pay for perfect information.
 - $EVPI = \text{Expected payoff with perfect information} - \text{Expected payoff without experimentation}.$
 - For Goferbroke Co., $EVPI = 242.5 - 100 = 142.5.$
- **Expected Value of Experimentation (EVE):**
 - The actual improvement in expected payoff from experimentation.
 - $EVE = \text{Expected payoff with experimentation} - \text{Expected payoff without experimentation}.$
 - For Goferbroke Co., $EVE = 123 - 100 = 23.$

Key Take away points

- Experimentation can improve decision-making by reducing uncertainty.
- Posterior probabilities are calculated using Bayes' theorem.
- The value of experimentation (EVE) should be compared to its cost to determine if it is worthwhile.
- Decision trees are a useful tool for analyzing decisions with experimentation.

Decision Tree for Experimentation

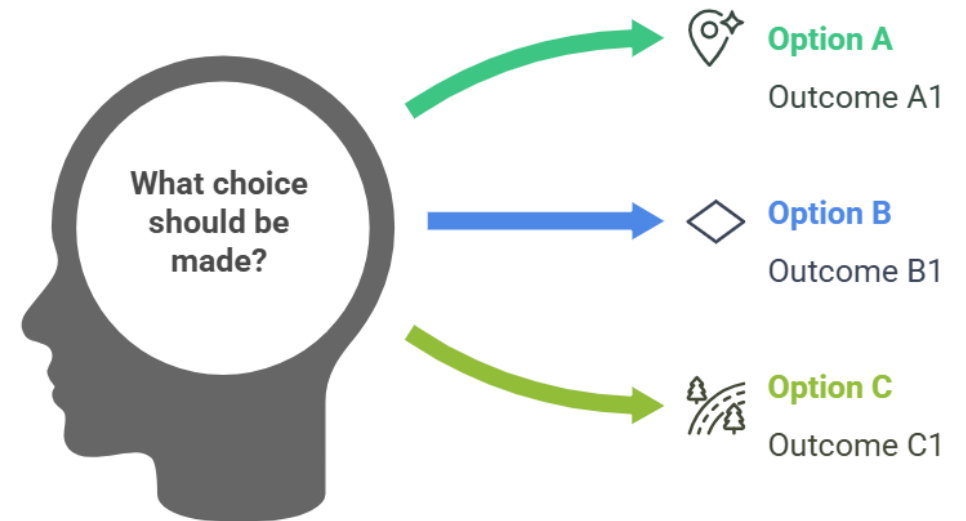
- A decision tree visually represents the decision-making process with experimentation.
- Nodes:
 - **Decision nodes:** Represent choices (e.g., conduct survey or not).
 - **Event nodes:** Represent random events (e.g., survey results, oil or dry).
- Example: Decision tree for Goferbroke Co. showing all possible paths and payoffs.

16.4 Decision Trees

Mapping Out Smart Choices

Why we use a decision tree?

- Shows decisions and Uncertainty in order
- Helps when we face multiple choices over time
- Combine options, chances, and payoffs.



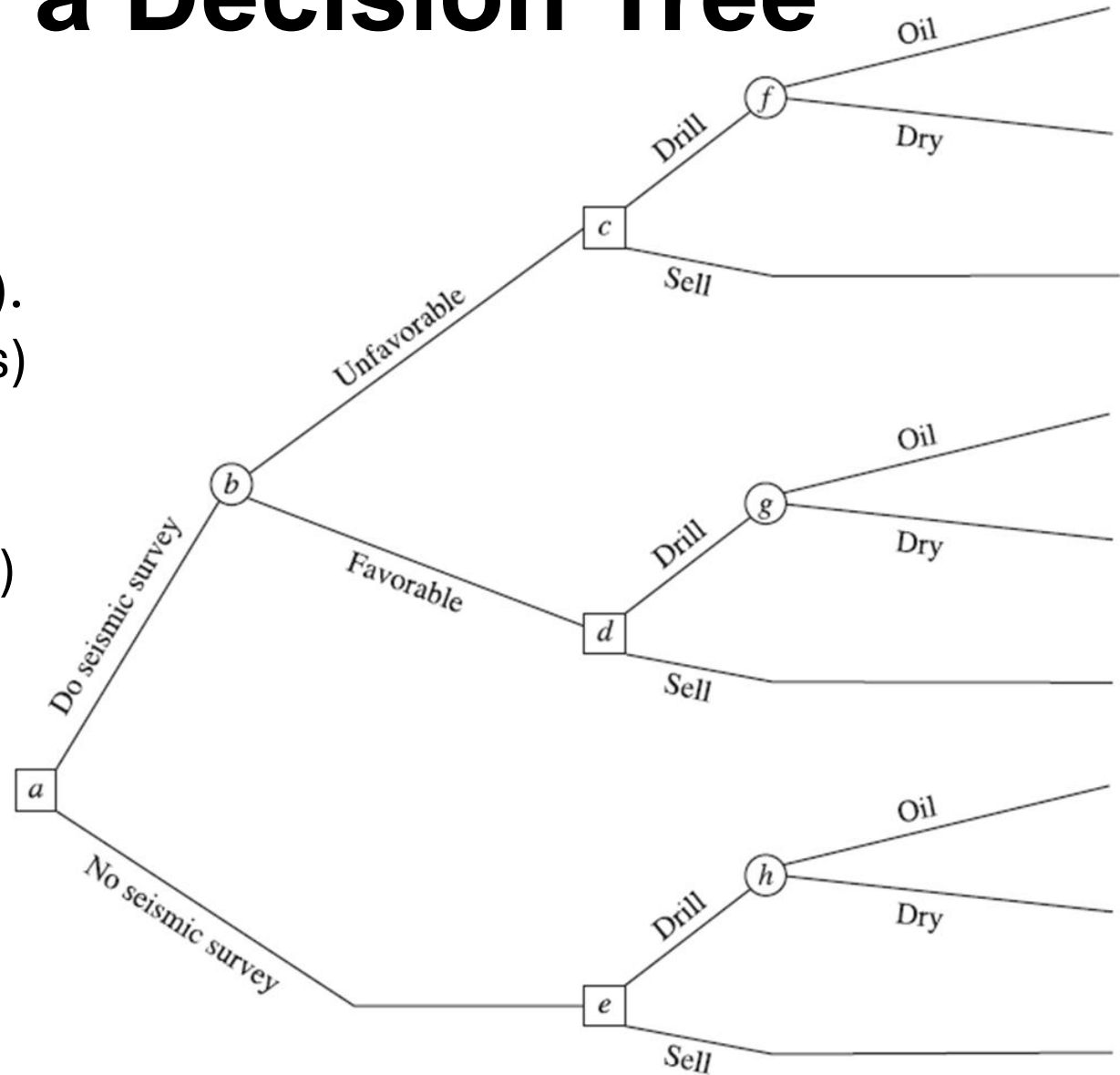
Goferbroke's two step Problem

- Goferbroke's Big Picture
- Decision 1: Do seismic survey (\$30k) or not?
- Decision 2: Drill (\$700k oil, -\$100k dry) or sell (\$90K)?
- Uncertainty: Oil (25%) or dry (75%), survey results

Building Blocks of a Decision Tree

Nodes:

- Square = Decision (you choose).
- Circle = Event (chance happens)
- Branches : Paths for choices or outcomes.
- Payoffs: Results at the end (in \$)



■ **FIGURE 16.4**

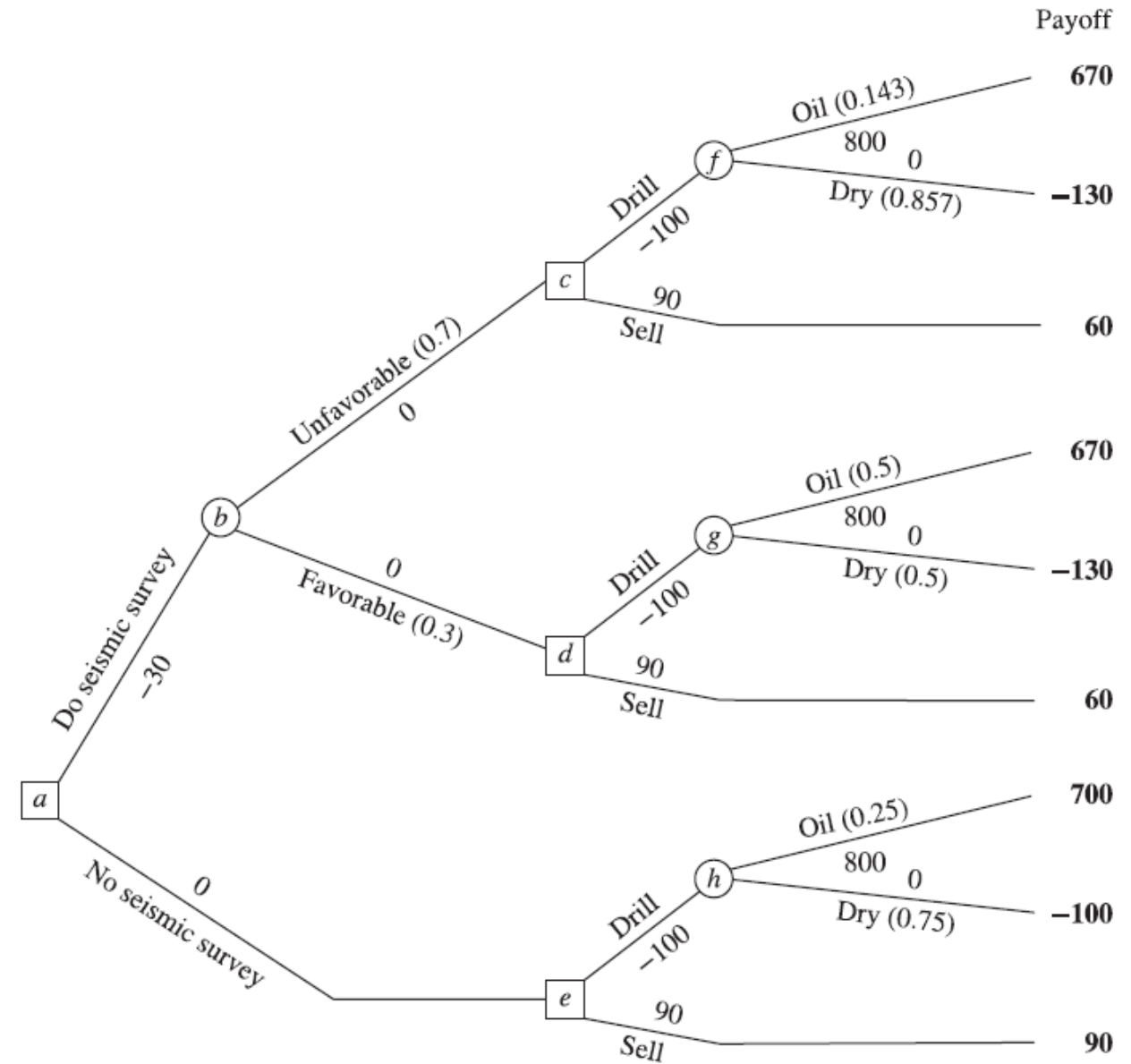
The decision tree (before including any numbers) for the full Goferbroke Co. problem.

Goferbroke's Decision Tree

- Start: Test or No Test?
- Test → FSS or USS → Drill or Sell.
- No Test → Drill or Sell.

Decision tree with probabilities of the random events and the payoffs

- Square = Decision (you choose). It indicates that the decision need to be made at the point in the process
- Circle = Event (chance happens). It indicates that a random event occur at that point.
- Branches : Paths for choices or outcomes.
- Payoffs: Results at the end (in \$)

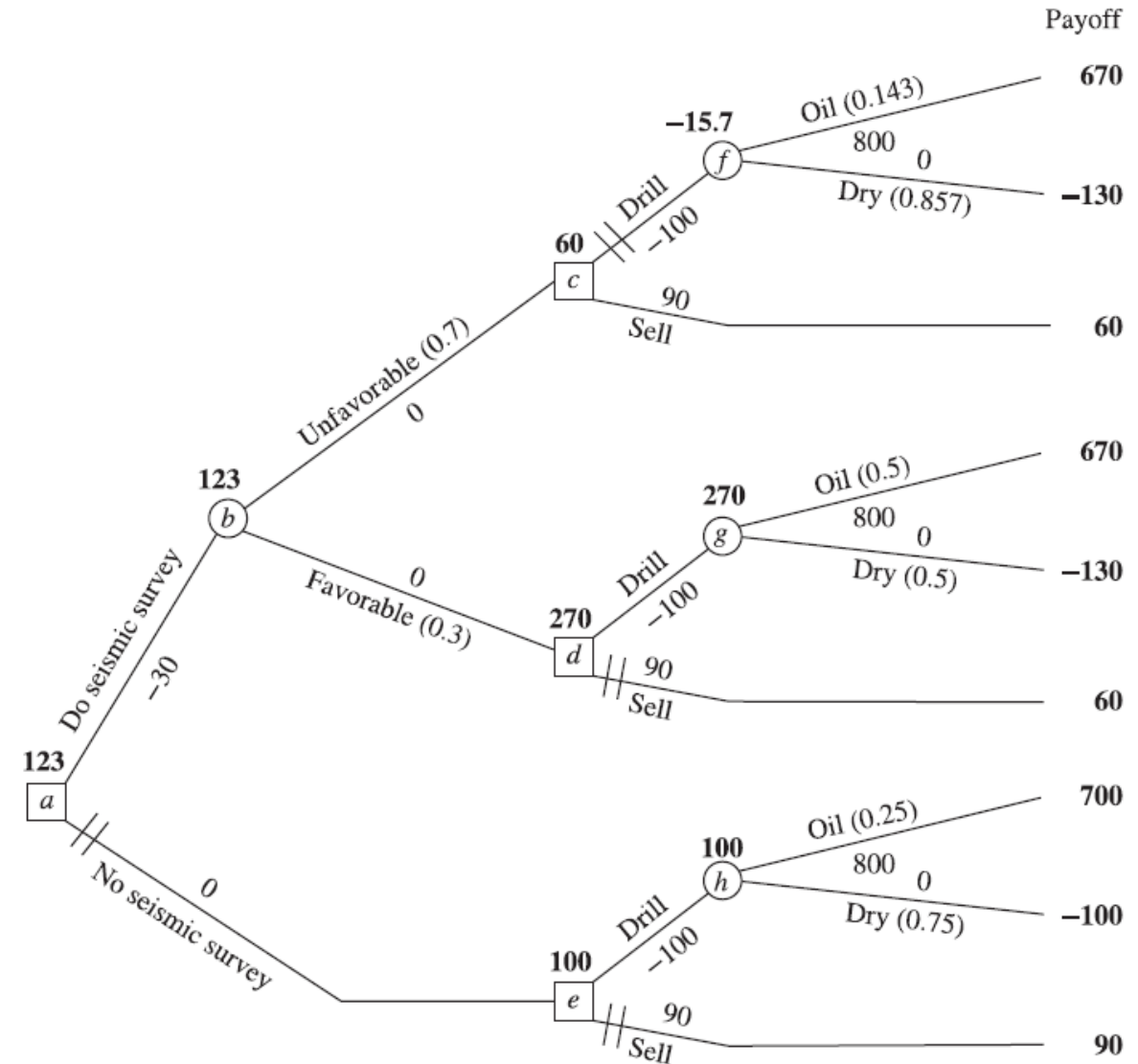


■ **FIGURE 16.5**

The decision tree in Fig. 16.4 after adding both the probabilities of random events and the payoffs.

Full Tree Overview

- Shows all paths and payoffs
- End values (in \$k):
- Test, FSS, Drill, Oil: 670.
- Test, USS, Sell: 60.
- No Test, Drill, Dry: -100.



■ **FIGURE 16.6**
The final decision tree that records the analysis for the full Goferbroke Co. problem when using monetary payoffs.

Analyzing the tree

- Backward Induction
 1. Start at the end (payoffs).
 2. Work back, picking best options.
 3. Average chance events.
- Goal: Maximize expected payoff.

Step 1 - End Choices (Pick at the Leaves)

- **FSS:** Drill ($0.5 \times 670 + 0.5 \times -130 = 270$), Sell (60).
 - Choose Drill: \$270k.
- **USS:** Drill ($1/7 \times 670 + 6/7 \times -130 = -15.7$), Sell (60).
 - Choose Sell: \$60k.
- **No Test:** Drill ($0.25 \times 700 + 0.75 \times -100 = 100$), Sell (90).
 - Choose Drill: \$100k.

Test or Not? Back to Start

- **Test:** $0.3 \times 270 + 0.7 \times 60 = 81 + 42 = \$123\text{k}.$
- **No Test:** \$100k (from Drill).
- **Winner:** Test, expect \$123k.

Optimal Policy (The Game Plan)

- Do the seismic survey (\$30k).
- If FSS: Drill (expect \$270k).
- If USS: Sell (expect \$60k).
- Overall: \$123k expected payoff.
 - Conduct the seismic survey.
 - If the result is unfavorable (USS), sell the land.
 - If the result is favorable (FSS), drill for oil.
- **Expected Payoff:** \$123,000 (including the cost of the seismic survey).

Checking the value (Was testing worth it?)

- No Test: \$100k (Drill).
- Test: \$153k (before cost), \$123k (after \$30k).
- Gain: $\$153k - \$100k = \$53k > \$30k$.

Advantages of Decision Trees

- **Visual Representation:-**
 - Easy to understand and communicate.
- **Structured Approach:-**
 - Helps break down complex decisions into manageable parts.
- **Flexibility:-**
 - Can handle sequential decisions and uncertainty.
- **Quantitative Analysis:-**
 - Allows for calculation of expected payoffs and optimal decisions.

Limitations of Decision Trees

- **Complexity:** Can become large and unwieldy for problems with many decisions and outcomes.
- **Subjectivity:** Probabilities and payoffs are often based on estimates.
- **Static Analysis:** Assumes decisions are made in a fixed sequence.

Real-World Application of Decision Tree

- **Business:** Investment decisions, product launches, marketing strategies.
- **Healthcare:** Treatment options, diagnostic testing.
- **Engineering:** Project management, risk analysis.
- **Finance:** Portfolio management, insurance decisions.

Summary

- Decision trees are a powerful tool for visualizing and analyzing sequential decision problems.
- They help identify optimal decisions by calculating expected payoffs.